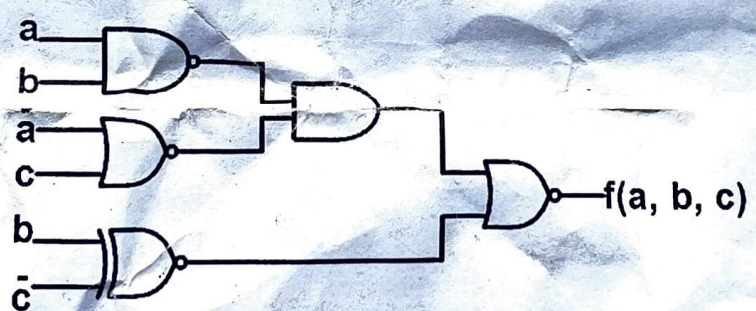
(Please write your Enrollment Number) Enrollment No. 8

MID-TERM EXAMINATION

(Course Name : BTECH ECE AI) (Semester -2nd)

(May, 2023)OFF LINE mode

Subject Code: BEC 104	Subject: Digital Electronics
Time : 1 ½Hours	Maximum Marks : 30
Note:Q. 1 is compulsory.	

Q1		(2.5	
	(a) Draw the Symbol and write the Truth Table of Universal Gates.		
	(b) Convert the Gray code 110101 to Binary form.		
	(c) Discuss between combinational and sequential circuit, with few example to justify your answer.		
	(d) What is a Flip-Flop? Give examples		
Q2	(Attempt any Two Parts) UNIT-1	(5,5)	
	(a)Draw the block diagram and write the Truth Table of Full Adder. Show all steps clearly.		
	(b) Draw a K-map for the following function of four variables and use it to reduce the function. How you will realize the reduced function using: (i) NAND gates (ii) NOR gates. $F = (A, B, C, D) = \sum (0, 2, 5, 6, 7, 8, 10)$		
	(c)Construct a 4x16 decoder with two 3x8 decoder.		
Q3	(Attempt any Two Parts) UNIT-2	(5,5)	
	(a)Analyze the combinational circuit, Show ALL steps.  		
	(b)How is SR Filp-Flop realized employing NOR gates. Explain using proper Table with ALL inputs.		
	(c) Implement a 16:1 MUX using 4:1 MUX		